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CLINICAL TRIALS AND OBSERVATIONS

Isatuximab, bortezomib, lenalidomide, and dexamethasone for multiple myeloma: dynamics of MRD negativity in the IMROZ study

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KEY POINTS

- **Isa-VRd/Isa-Rd led to higher MRD negativity and MRD-negative CR rates after induction vs VRd/Rd, with rates increasing during maintenance.**
- **Isa-VRd/Isa-Rd was also associated with a lower rate of conversion from MRD negativity to MRD positivity vs VRd/Rd.**

Quadruplet therapy with Isa-VRd (isatuximab, Velcade [bortezomib], Revlimid [lenalidomide], and dexamethasone) followed by Isa-Rd in the randomized phase 3 IMROZ study provided a significant progression-free survival benefit to transplant-ineligible patients with newly diagnosed multiple myeloma (NDMM). In the primary analysis, more Isa-VRd/Isa-Rd–treated patients achieved minimal residual disease (MRD) negativity and MRD-negative complete response (CR) at any time point than patients receiving VRd followed by Rd. Here, we report landmark analysis results of MRD negativity over time and its impact on clinical outcomes in IMROZ. Treatment with Isa-VRd/Isa-Rd led to deeper responses, with higher rates of MRD negativity and MRD-negative CR at the end of induction and during maintenance vs VRd/Rd, up to 60 months of follow-up. Benefit with Isa-VRd/Isa-Rd was observed across key patient subgroups, including older (>70 years) and frail patients. Time to progression (TTP) was significantly prolonged with Isa-VRd/Isa-Rd vs VRd/Rd in patients who converted from MRD negative to MRD positive at any time point. In a landmark analysis of the time of conversion to MRD positivity, TTP in patients who converted from MRD negative at the end of induction to MRD positive also favored Isa-VRd/Isa-Rd. Evaluating MRD status of patients at >1 time point may therefore be useful to support decisions on treatment selection and treatment continuation/discontinuation. Our findings on the degree of MRD negativity and MRD-negative CR benefit achieved during induction and maintenance by patients receiving Isa-VRd/Isa-Rd vs VRd/Rd extend the IMROZ primary analyses and further support Isa-VRd as standard of care for frontline treatment of transplant-ineligible patients with NDMM. This trial was registered at www.clinicaltrials.gov as #NCT03319667.

Introduction

Minimal residual disease (MRD) status affects outcomes in both transplant-eligible and -ineligible patients with newly diagnosed multiple myeloma (NDMM) as well as patients with relapsed/refractory MM.[1-3] Patients with NDMM who achieve MRD negativity demonstrate longer progression-free survival (PFS) and overall survival than MRD-positive patients,[2,3] with sustained MRD negativity associated with superior outcomes and loss of MRD negativity associated with worse prognosis.[4,5] Some studies have also demonstrated potential survival benefits of early MRD negativity achievement, and, although others have not,[3,6] the growing evidence supporting the importance of achieving MRD negativity to patient outcomes has contributed to MRD status being actively evaluated as a surrogate end point for PFS in patients with NDMM.[2-5] In April 2024, the Oncologic Drugs Advisory Committee voted unanimously to accept MRD negativity as a reliable early end point for clinical studies in MM.[7,8] Given that MRD negativity plays an increasingly important role as an early readout of efficacy with the advent of additional effective therapies, more insight into its dynamics and clinical relevance may help optimize treatment strategies and support decisions on treatment selection and continuation.

Results from randomized, phase 3 studies with anti-CD38 quadruplet regimens that include a proteasome inhibitor and an immunomodulatory drug demonstrated improved outcomes for patients with NDMM who were not eligible for autologous stem cell transplantation (ASCT) as well as patients with NDMM who were ASCT candidates, compared with those with conventional therapies.[9-18] Isatuximab (Isa) is an immunoglobulin G1 monoclonal antibody that targets a specific epitope of human CD38 on MM cells, inducing myeloma cell death by multiple modes of action.[19] Based on the results of the randomized, phase 3 IMROZ study and supported by the outcomes observed in the randomized, phase 3 BENEFIT study, Isa was recently approved in combination with VRd (Velcade [bortezomib], Revlimid [lenalidomide], and dexamethasone) for the frontline treatment of transplant-ineligible patients with NDMM aged ≤80 years in the United States and the European Union, among others.[20,21] In the IMROZ study (ClinicalTrials.gov identifier: NCT03319667), transplant-ineligible patients with NDMM achieved significantly improved PFS (hazard ratio [HR], 0.60; 98.5% confidence interval [CI], 0.41-0.88; P < .001) and deep, sustained responses with Isa-VRd quadruplet therapy followed by Isa-Rd (Isa-VRd/Isa-Rd) compared with those with VRd triplet therapy followed by Rd (VRd/Rd).[10] In the primary analysis, at a median follow-up of 59.7 months, more patients treated with Isa-VRd/Isa-Rd than with VRd/Rd in the intent-to-treat (ITT) population achieved MRD negativity (at a 10^{-5} sensitivity threshold) (58.1% vs 43.6%), MRD negativity with complete response (CR; 55.5% vs 40.9%), and 12-month sustained MRD negativity (46.8% vs 24.3%).[10] Furthermore, in the randomized, phase 3 BENEFIT study, an Isa-VRd regimen with a weekly bortezomib schedule demonstrated significant benefits in transplant-ineligible patients with NDMM, with MRD negativity rates (10^{-5} sensitivity threshold, primary end point) of 53% for Isa-VRd vs 26% for Isa-Rd at 18 months from randomization (odds ratio [OR], 3.16; 95% CI, 1.89-5.28; P < .0001).[11]

Here, we report results from analyses conducted to assess the dynamics of MRD negativity over time, sustained MRD negativity for ≥24 months, and its impact on clinical outcomes in patients treated with Isa-VRd induction followed by Isa-Rd maintenance vs VRd followed by Rd in the IMROZ study. In addition, we build on the MRD results from the interim analysis of the IMROZ study by investigating the impact of achieving early MRD negativity and its implications for PFS and time to next treatment as well as the impact of MRD negativity on key subgroups of interest including frail patients.

Methods

Trial design, randomization, and treatment

Trial design, patient inclusion/exclusion criteria, randomization, and treatment in the IMROZ study have been reported in detail previously.[9] Briefly, IMROZ was a global, randomized, open-label, phase 3 trial conducted in 21 countries. The trial enrolled patients aged ≥18 years with symptomatic, previously untreated MM and measurable disease who were not eligible to undergo ASCT due to an age of ≥65 years or due to coexisting conditions. The key exclusion criteria were an age of >80 years, an Eastern Cooperative Oncology Group performance status score of >2, or an estimated glomerular filtration rate (eGFR) of <30 mL/min per 1.73 m^2 of body surface area at screening. Patients were randomly assigned in a 3:2 ratio to receive either Isa-VRd/Isa-Rd or VRd/Rd. Randomization was stratified according to country (not China vs China), age (<70 vs ≥70 years), and Revised International Staging System disease stage (stage I or II vs III vs not classified). The independent ethics committee or institutional review board at each study center approved the trial protocol, and the trial was conducted in accordance with the principles of the Declaration of Helsinki and the guidelines for Good Clinical Practice of the International Council for Harmonisation. All patients provided informed consent.

The patients received 4 6-week induction cycles with Isa-VRd or VRd followed by 4-week cycles of continuous maintenance treatment with Isa-Rd (in the Isa-VRd/Isa-Rd arm) or Rd (in the VRd/Rd arm) until disease progression, an unacceptable adverse event, or other discontinuation criteria.

During the induction phase, patients in the Isa-VRd arm received IV Isa at 10 mg/kg of body weight once weekly in cycle 1 and then every 2 weeks in subsequent cycles. All patients received VRd, which consisted of subcutaneous bortezomib (1.3 mg/m^2 on days 1, 4, 8, 11, 22, 25, 29, and 32), oral lenalidomide (25 mg/day [or 10 mg/day if the eGFR was 30 to <60 mL/min per 1.73 m^2] on days 1-14 and 22-35), and oral or IV dexamethasone (20 mg/day on days 1, 2, 4, 5, 8, 9, 11, 12, 15, 22, 23, 25, 26, 29, 30, 32, and 33 [or on days 1, 4, 8, 11, 15, 22, 25, 29, and 32 for patients aged ≥75 years]).

During continuous maintenance treatment, patients in the Isa-VRd arm followed by Isa-Rd arm received IV Isa at 10 mg/kg every 2 weeks, with monthly administration starting with cycle 18. All patients received an Rd regimen that consisted of oral lenalidomide (25 mg/day [or 10 mg/day if the eGFR was 30 to <60 mL/min per 1.73 m^2] on days 1-21) and dexamethasone (20 mg/day once weekly). Crossover from the Rd regimen to the Isa-Rd regimen was allowed during continuous treatment at the investigator's discretion, in case of confirmed progression, and occurred for 25 patients.[10]

Assessments

MRD negativity was assessed by next-generation sequencing (NGS) using the Food and Drug Administration–approved clonoSEQ Assay version 2.0 (Adaptive Biotechnologies), with results reported at the 10^{-5} and 10^{-6} sensitivity thresholds. Bone marrow aspirates for MRD assessments were obtained at baseline, at the end of induction (month 6), and during maintenance (months 12, 18, 24, and then yearly) from patients with a very good partial response (VGPR) or better. VGPR status was used as a threshold for bone marrow sampling to reduce unnecessary invasive procedures in patients with PR or less who were unlikely to have achieved MRD negativity. The number of samples per patient was therefore dynamic and not fixed; per protocol, samples were collected for patients reaching ≥VGPR until the patient progressed, withdrew study consent, or died. An average of 4.7 samples for the Isa-VRd/Isa-Rd arm and 4.1 samples for the VRd/Rd arm were collected per patient at the time of the cutoff date. All MRD assessment time points were ±3 months, and MRD assessments in the ITT population counted patients with no or undetermined (ie, insufficient number of cells assayed to confidently assign MRD status at the given sensitivity threshold) MRD assessments as MRD positive. Furthermore, patients who crossed over from the Rd to the Isa-Rd regimen were not included in the analyses for MRD dynamics over time. Sustained MRD negativity for ≥12, ≥24, or ≥36 months was defined as the proportion of patients who maintained MRD negativity, confirmed ≥12, ≥24, or ≥36 months apart, with no MRD-positive test in between. Exploratory analyses included MRD analyses by NGS at the 10^{-6} sensitivity threshold.

Central laboratories performed disease and MRD assessments. The members of an independent review committee, unaware of treatment assignment, determined disease response or progression based on efficacy data (assessed at a central laboratory), bone marrow evaluations for plasma cell infiltration (assessed at a local laboratory), and centrally reviewed imaging, according to the International Myeloma Working Group criteria.

Statistical analysis

To mitigate lead-time bias inherent to MRD conversion analyses, outcomes were evaluated using landmark analyses starting from the time of MRD status conversion. However, it is important to acknowledge that other potential biases may remain in this type of analysis. The landmark analysis evaluated the proportion of patients (n [%]) achieving a specific MRD status (eg, MRD negativity) at a predefined time point (the "landmark"). Inclusion at the landmark time required patients to be alive and have an available MRD assessment.

Median PFS from the start of treatment and corresponding CIs were calculated by the Kaplan-Meier method. A Cox proportional hazards model was used to obtain HR estimates. Continuous variables were summarized using descriptive statistics, and categorical and ordinal variables were summarized as frequencies and percentages. ORs were calculated using unconditional maximum likelihood estimation and 95% CIs using normal approximation. All the analyses were performed in SAS Viya version 4.

Results

In the IMROZ study, a total of 446 patients (ITT population) were randomized to receive Isa-VRd/Isa-Rd (n = 265) or VRd/Rd (n = 181). The dynamics of MRD negativity in treated patients were evaluated by NGS using 1610 MRD assessments from 361 patients (Isa-VRd/Isa-Rd, n = 223; VRd/Rd, n = 138). The median duration of MRD assessments was 4 years (maximum: 5.4 years) for Isa-VRd/Isa-Rd and 3 years (maximum: 5.5 years) for VRd/Rd patients (supplemental Table 1, available on the *Blood* website).

MRD negativity

An analysis of the cumulative MRD negativity rates (at the 10^{-5} and 10^{-6} sensitivity thresholds) in the ITT population showed that, although the median time from first dose to initial MRD negativity in the evaluable population was 6.5 months in both arms, more patients achieved MRD negativity with Isa-VRd/Isa-Rd than with VRd/Rd at later time points throughout the study (Figure 1). MRD negativity with 10^{-6} sensitivity was not evaluable in all samples analyzed, but the success rate was ~87%. In a landmark analysis at 6 months among patients who were MRD negative, there was no statistically significant difference in PFS between Isa-VRd/Isa-Rd and VRd/Rd, (at 10^{-5}, HR, 0.562; 95% CI, 0.296-1.068; P = .0784; Figure 2).

The evaluation of MRD negativity rates by study time point (up to 60 months) showed that a higher proportion of patients were MRD negative in the Isa-VRd/Isa-Rd arm than in the VRd/Rd arm throughout the study (Figure 3). The significant

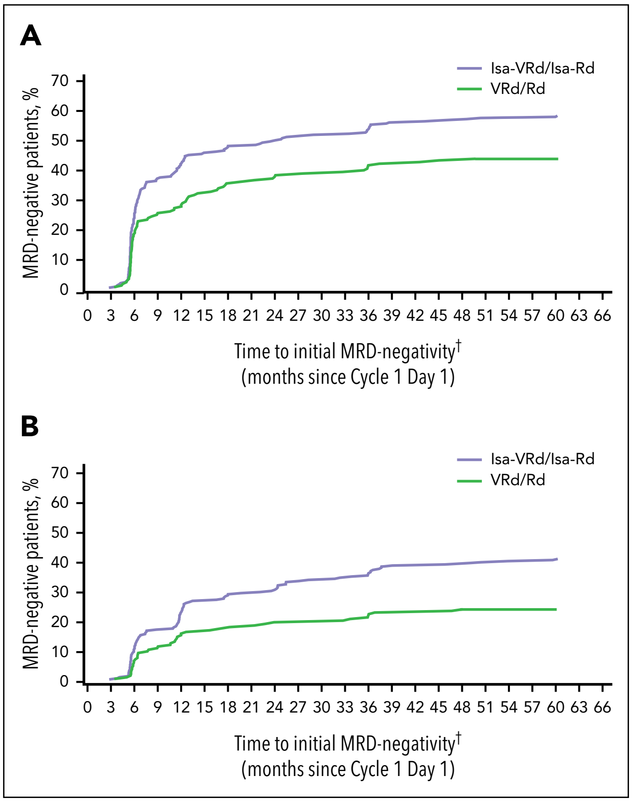
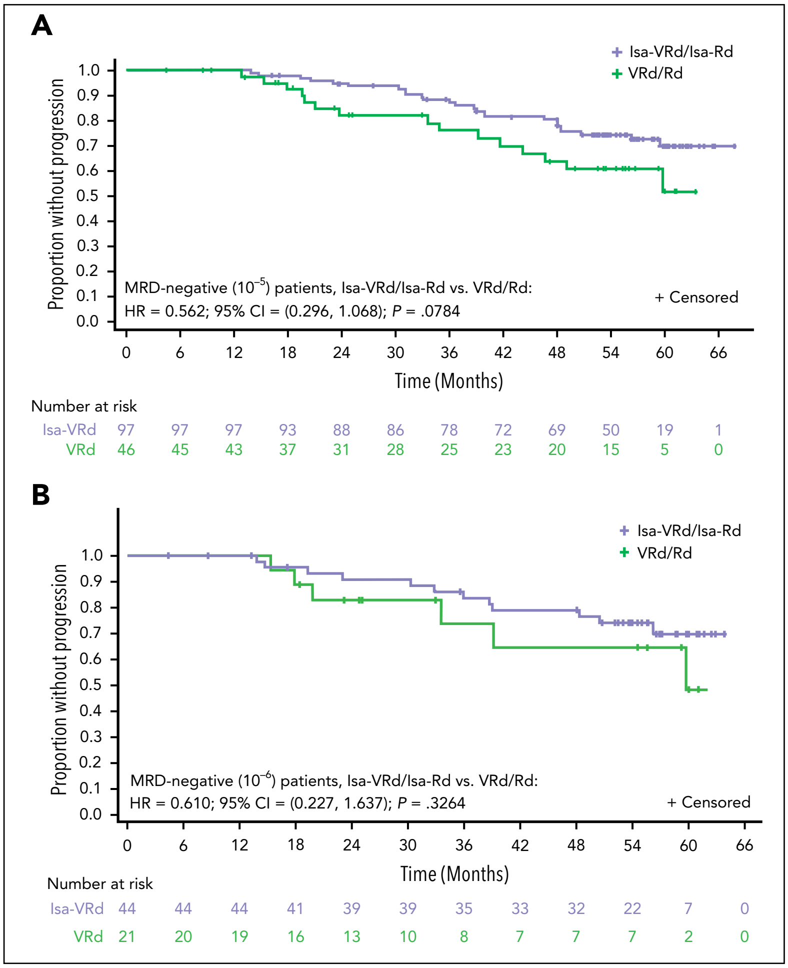


Figure 1. Cumulative MRD negativity rates in the ITT population. MRD negativity rates at 10^{-5} sensitivity threshold (A) and 10^{-6} sensitivity threshold (B). All time points were calculated using the number of patients who achieved initial MRD negativity at that time point divided by the ITT (Isa-VRd/Isa-Rd, n = 265; VRd/Rd, n = 181); the rate is cumulative over time. †The last MRD-negative result in the VRd/Rd arm occurred at 48 months; beyond this, the cumulative rate did not increase and can be considered a flat line.

Figure 2. PFS in patients who were MRD negative at 6 months (induction phase). (A) PFS in MRD-negative patients at the 10^{-5} sensitivity threshold (P = .0784). (B) PFS in MRD-negative patients at the 10^{-6} sensitivity threshold (P = .3264).



benefit observed with Isa-VRd/Isa-Rd vs VRd/Rd in the maintenance phase was maintained over time, with MRD negativity rates (10^{-5}) of 54.0% vs 39.2% at 12 months (OR, 1.81; 95% CI, 1.11-2.98) and up to 76.1% vs 40.0% at 60 months (OR, 4.59; 95% CI, 1.34-17.04), respectively.

Patient subgroups

Patients in key subgroups defined by age, baseline eGFR levels, Revised International Staging System disease stage, presence of a plasmacytoma (ie, extramedullary disease or paramedullary disease), high-risk cytogenetics status (defined as the presence of at least one of the following: del(17p) and/or t(4;14) and/or t(14;16)), 1q21^{+} status, and frailty (with frailty defined per simplified International Myeloma Working Group scores of ≥2)[22] were more likely to achieve MRD negativity in the Isa-VRd followed by Isa-Rd arm than patients treated in the VRd followed by Rd arm (at 10^{-5} and 10^{-6} sensitivity thresholds; supplemental Figure 1). Moreover, an analysis of MRD negativity rates by study time point in patient subgroups defined by

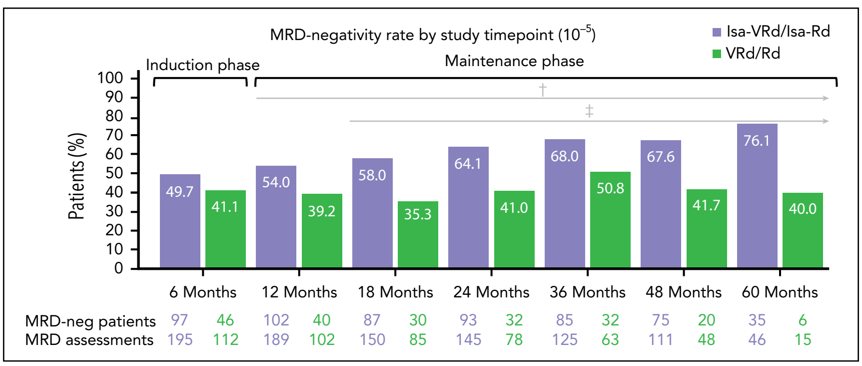


Figure 3. Landmark MRD negativity analyses during induction and maintenance phases. MRD negativity at the induction phase and at various time points during the maintenance phase. †Bortezomib was dropped and patients received Isa-Rd and Rd, respectively. ‡Patients received monthly isatuximab. MRD-neg, minimal residual disease–negative.

differences in age (≤70 years and >70 years) and frailty (frail and nonfrail) showed a consistent benefit with Isa-VRd/Isa-Rd across subgroups at each study time point in the maintenance phase (supplemental Figure 2A-B).

Conversions of MRD status

Assessing conversion from MRD positivity at the induction phase to MRD negativity at various time points during maintenance revealed that a higher proportion of patients in the Isa-VRd/Isa-Rd arm than in the VRd/Rd arm converted from MRD positivity to MRD negativity during maintenance (36.1% vs 18.0% at 24 months and 48.2% vs 33.3% at 36 months), with an increasing conversion rate with Isa-VRd/Isa-Rd vs VRd/Rd over the maintenance phase (supplemental Figure 3).

Evaluating conversion from MRD negativity at the induction phase to MRD positivity in the maintenance phase showed that a far lower proportion of MRD-negative patients became MRD positive at various time points during maintenance in the Isa-VRd/Isa-Rd arm than in the VRd/Rd arm (5.6% vs 26.9% at 24 months and 12.3% vs 34.8% at 36 months; Figure 4). In a landmark analysis, patients who converted to MRD positivity during maintenance after an MRD-negative assessment at the induction phase demonstrated a significant time-to-progression (TTP) benefit with Isa-VRd/Isa-Rd vs VRd/Rd (HR, 0.236; 95% CI, 0.089-0.624; $P = .0036$; Figure 5A). In addition, in a landmark analysis from conversion to MRD positivity at any time point following MRD negativity, the evaluation of TTP after conversion from MRD negative to MRD positive at any time point during the study showed that patients who converted to MRD positivity derived a superior benefit from Isa-VRd followed by Isa-Rd compared with that from VRd followed by Rd, with a significantly longer TTP (HR, 0.275; 95% CI, 0.114-0.664; $P = .0041$; Figure 5B). The baseline characteristics of the subgroup of patients who maintained/lost MRD negativity during the maintenance phase are presented in supplemental Table 2.

MRD-negative CR

The evaluation of PFS in MRD-negative patients who achieved a CR showed a significant PFS benefit with Isa-VRd/Isa-Rd compared with that with VRd/Rd in the ITT population (HR, 0.539; 95% CI, 0.304-0.953; $P = .0336$; 10^{-5} sensitivity; Figure 6A). In a landmark analysis of MRD negativity rates (10^{-5} sensitivity threshold) at different study time points in patients with CR, a consistently higher proportion of patients reached an MRD-negative CR with Isa-VRd/Isa-Rd vs VRd/Rd over time up to 60 months of observation (73.9% vs 40.0%; Figure 6B). Isa-VRd/Isa-Rd demonstrated no significant difference in PFS over VRd/Rd for CR patients with MRD negativity at the 10^{-6} sensitivity threshold (supplemental Figure 4). Furthermore, subgroup analyses showed a higher likelihood of achieving an MRD-negative CR with Isa-VRd/Isa-Rd than with VRd/Rd across key patient subgroups in the ITT population (at the 10^{-5} and 10^{-6} sensitivity thresholds) (supplemental Figure 5).

Sustained MRD negativity

The rates of sustained MRD negativity for ≥12, ≥24, and ≥36 months (at 10^{-5} and 10^{-6}) in the ITT population were approximately twofold to threefold higher with Isa-VRd/Isa-Rd than with VRd/Rd (at 10^{-5}: sustained for ≥12 months, 46.8% vs 24.3%; OR, 2.73; 95% CI, 1.80-4.14; sustained for ≥24 months, 35.8% vs 13.3%; OR, 3.65; 95% CI, 2.22-6.03; Figure 7). The evaluation of sustained MRD negativity for ≥36 months in this study was affected by immature data and follow-up will be continued.

The PFS for ITT patients with sustained MRD negativity for ≥24 months is shown in supplemental Figure 6. Patients were more likely to achieve sustained MRD negativity with Isa-VRd/Isa-Rd than with VRd/Rd; however, once patients achieved sustained MRD negativity for ≥24 months, PFS was similar in both treatment arms.

Subgroup analyses for 12-month sustained MRD negativity showed that patients across key subgroups were more likely to achieve sustained MRD negativity in the Isa-VRd/Isa-Rd arm than in the VRd/Rd arm (supplemental Figure 7).

Discussion

The achievement of MRD negativity in patients with MM is being recognized as a key efficacy end point correlated with longer PFS and better clinical outcomes across different patient populations.[8,23,24] To our knowledge, we have conducted the first detailed analyses of MRD dynamics in transplant-ineligible patients with NDMM in the IMROZ study, assessing the effects of anti-CD38 Isa-based quadruplet therapy over ~60 months. Isa-VRd followed by Isa-Rd led to deeper responses over time, with higher rates of MRD negativity and MRD-negative CR at both the end of the induction phase and during maintenance compared with those with the VRd/Rd arm. These findings are consistent with the IMROZ primary analysis, which reported MRD negativity (10^{-5} at any time

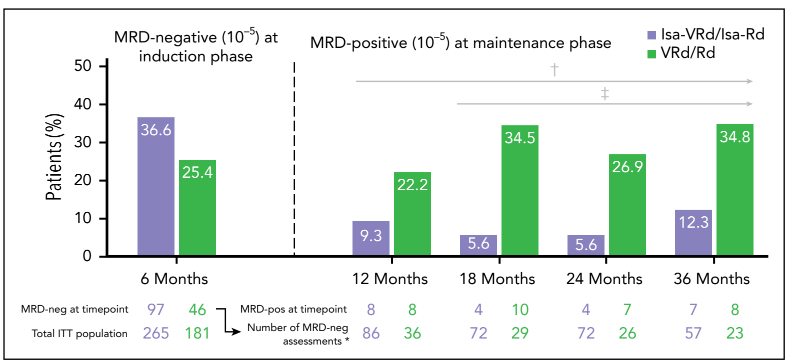


Figure 4. Landmark analysis: conversion from MRD negativity at induction phase to MRD positivity during maintenance phase. Conversion rates of MRD negativity at the induction phase to MRD positivity at various time points during the maintenance phase. Only patients who achieved MRD negativity (left side of the graph) during induction are depicted here and were evaluated for conversion to MRD positivity at any point during the maintenance phase (right side of the graph, shown from 12 through 36 months). *The number of assessments from MRD-negative patients at induction who had repeat MRD assessment at the given time point. †Bortezomib was dropped and patients received Isa-Rd and Rd, respectively. ‡Patients received monthly isatuximab. MRD-neg, minimal residual disease–negative; MRD-pos, minimal residual disease–positive.

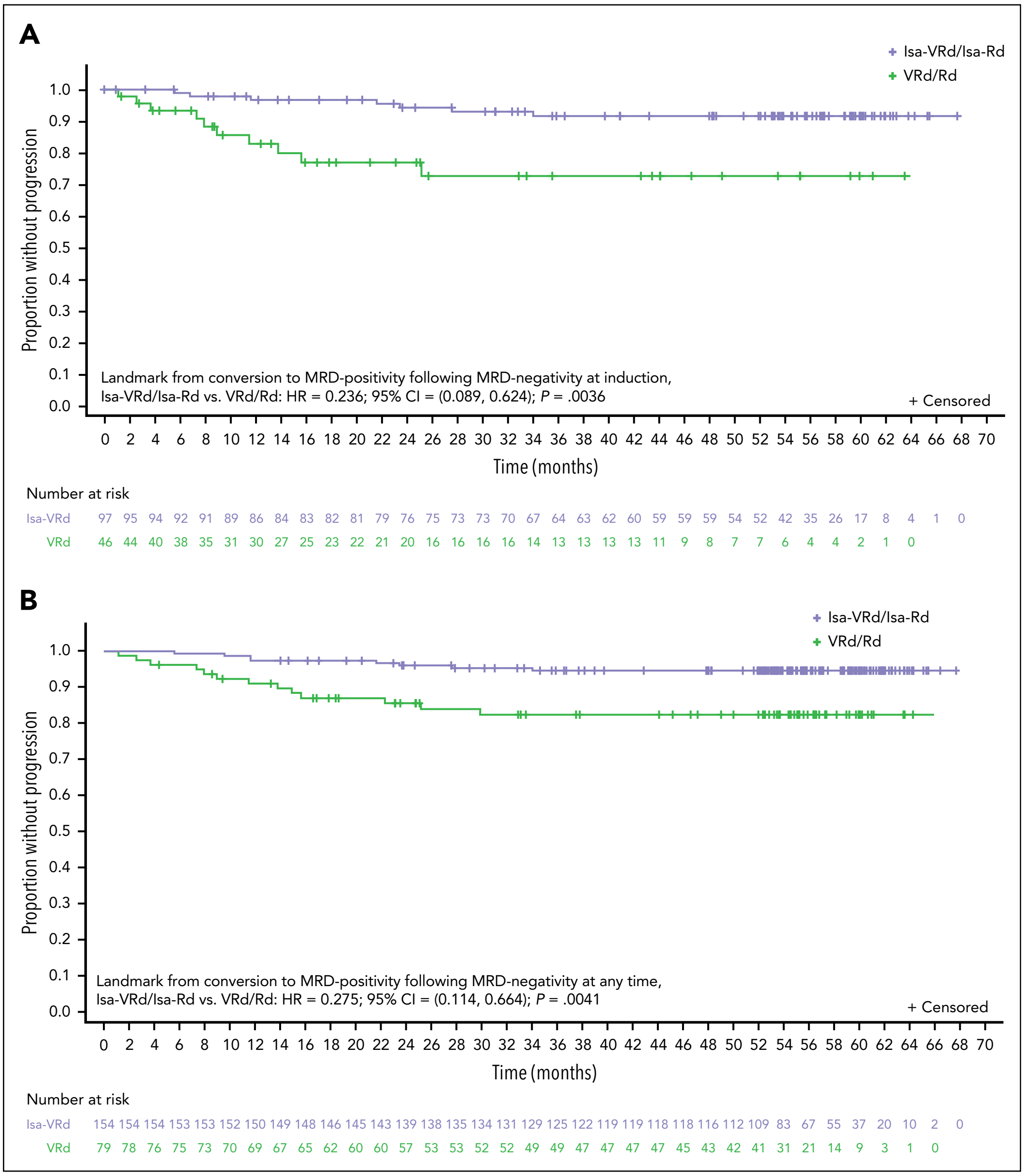


Figure 5. TTP. (A) After conversion from MRD negativity at the induction phase to MRD positivity during the maintenance phase (P = .0036). (B) After conversion from MRD negativity to MRD positivity at any time point (P = .0041).

point) of 58.1% for Isa-VRd/Isa-Rd vs 43.6% for VRd/Rd in the ITT population.[9] They are also in line with the high MRD negativity rates achieved with another Isa quadruplet regimen (Isa-KRd [isatuximab plus Kyprolis (carfilzomib), Revlimid (lenalidomide), and dexamethasone]) in the phase 3 MIDAS study in transplant-eligible patients with NDMM.[25] In this analysis, Isa-VRd/Isa-Rd demonstrated benefit across key patient subgroups, including older (>70 years) and frail patients, with higher rates of MRD negativity vs VRd/Rd at all time points evaluated in the landmark analyses performed up to 60 months of follow-up. It should be noted that some patients did not achieve MRD negativity until several years after treatment in the IMROZ study. Further investigation into the characteristics of this population would be of interest.

The deeper responses achieved throughout treatment were associated with improved outcomes, such as significantly improved PFS for MRD-negative patients achieving CR (with a 10^{-5} sensitivity threshold; Figure 6A), along with a significant prolongation of TTP in favor of Isa-VRd/Isa-Rd vs VRd/Rd among patients who converted from MRD negative to MRD positive during the study (Figure 5B). Importantly, a substantially lower rate of conversion from MRD negative to MRD positive was observed over time in the Isa-VRd/Isa-Rd arm vs the VRd/Rd arm. TTP in patients who converted from MRD negative to MRD positive at the end of the induction phase also favored Isa-VRd/Isa-Rd (Figure 5A), pointing to a lasting benefit over time from MRD negativity and disease control reached in the earlier phase of treatment with Isa-VRd.

The higher rates of sustained MRD negativity for ≥12 months and ≥24 months achieved in IMROZ with Isa-VRd/Isa-Rd vs VRd/Rd in the ITT population (10^{-5}, 46.8% vs 24.3% and 35.8%

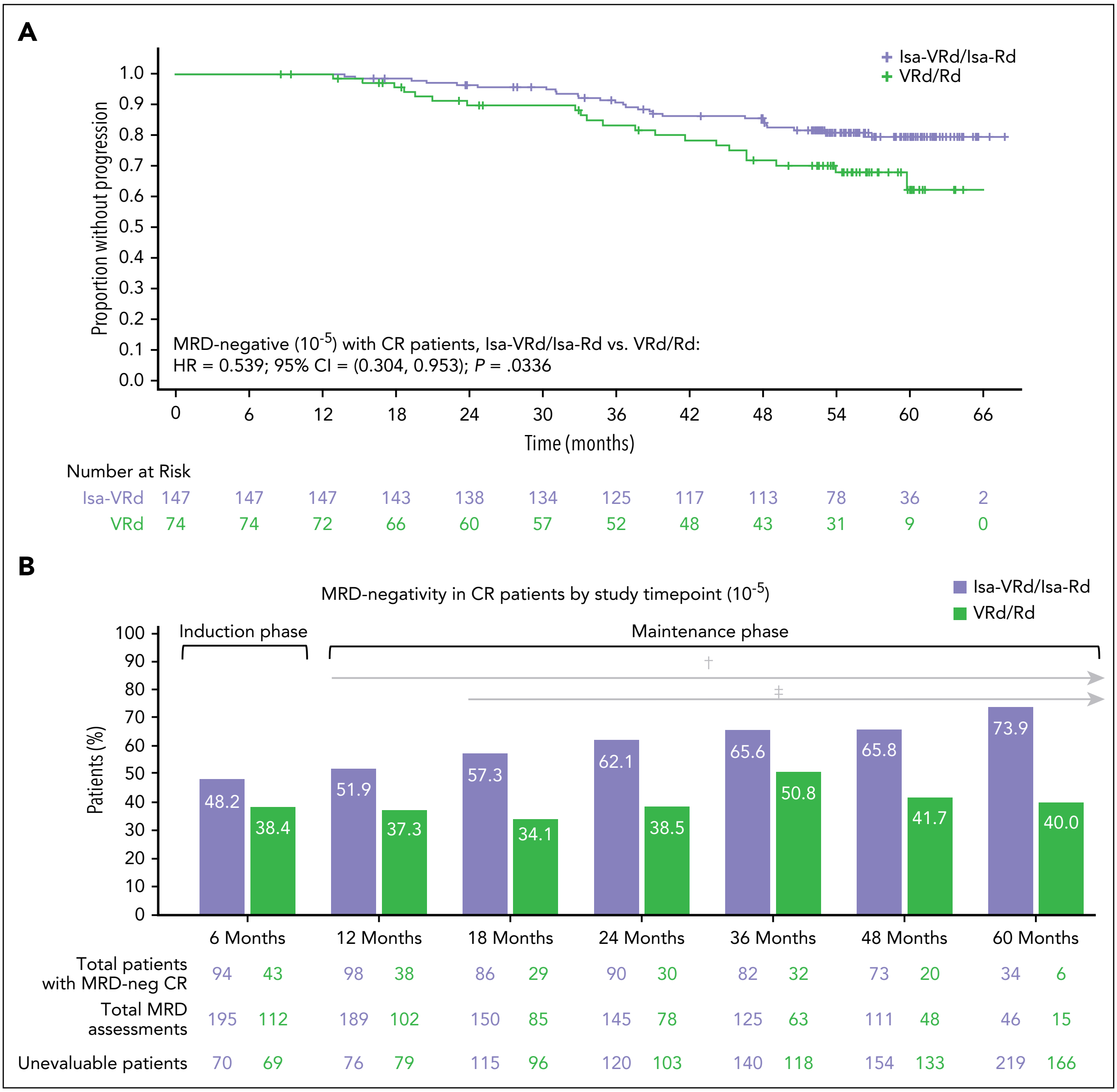


Figure 6. PFS for MRD-negative CR patients and MRD negativity rates over time. (A) PFS for MRD-negative CR patients (10^{-5}) in the ITT population (*P* = .0336). MRD negativity was defined as a patient having at least 1 MRD-negative assessment at any time point. Patients were MRD negative and achieved CR at any point in the study. (B) Landmark MRD negativity rates (10^{-5}) at different time points in patients who achieved CR. †Bortezomib was dropped and patients received Isa-Rd and Rd, respectively. ‡Patients received monthly isatuximab. MRD-neg, minimal residual disease–negative.

vs 13.3%, respectively) demonstrate the prolonged benefit obtained by adding Isa to standard VRd during induction and maintenance treatment. The BENEFIT study further demonstrated a consistent MRD benefit across subgroups with an Isa-VRd quadruplet regimen compared with that with the triplet regimen Isa-Rd. Notably, in IMROZ, MRD negativity was maintained in the Isa-VRd/Isa-Rd arm even after bortezomib withdrawal at 6 months and continued Isa-Rd treatment. The MRD benefit is maintained during the maintenance phase in the Isa-VRd/Isa-Rd arm, which suggests a potential remodeling of the tumor microenvironment to be less favorable for myeloma growth in an earlier phase of therapy, despite bortezomib withdrawal. Additional studies will be needed to test this hypothesis.

Nonetheless, it is important to consider the potential role bortezomib played in the results reported here. Bortezomib-mediated inhibition of proteasome activity provides a unique mechanism of action that contributes to the synergistic effect on MRD negativity achieved with the quadruplet regimen in this study. Several areas require further evaluation. First, additional studies are needed to understand whether achieving MRD negativity involves key changes to immune cells within the tumor microenvironment. Second, it is also possible that bortezomib discontinuation during the maintenance phase removes selective pressures on myeloma cells, allowing some patients to become MRD positive. More research is needed to understand additional risk factors underlying the loss of MRD negativity.

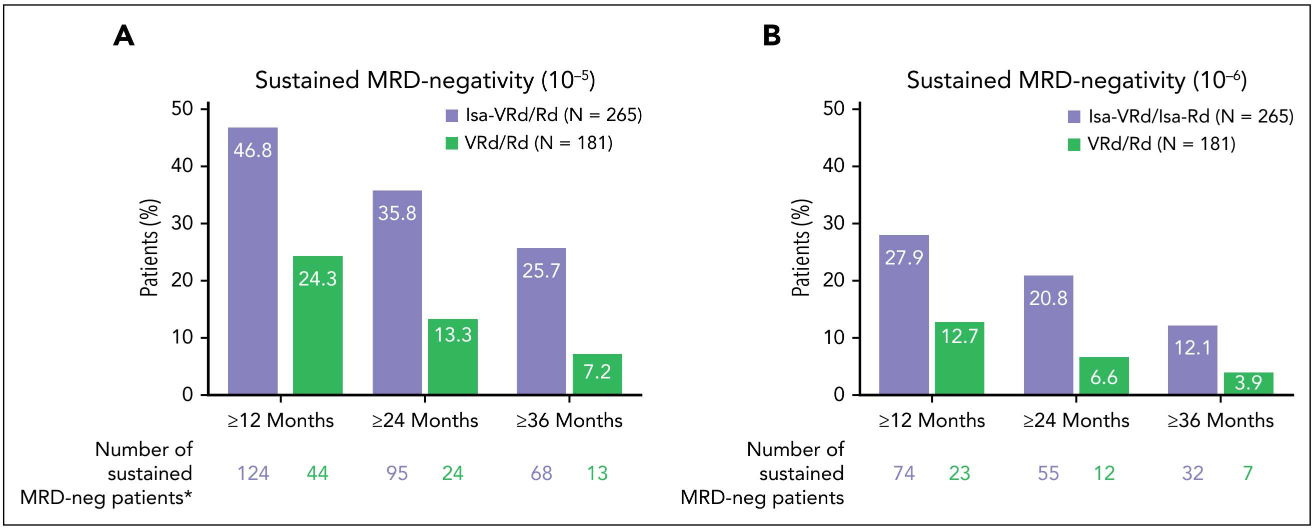


Figure 7. Sustained MRD negativity rates. MRD negativity at 10^{-5} sensitivity threshold (A) and 10^{-6} sensitivity threshold (B) in the ITT population. *A total of 13, 16, and 39 MRD-positive patients, who were MRD negative (10^{-5}) at some point on study, were missing in sustained MRD negativity assessments at ≥12, ≥24, and ≥36 months in the Isa-VRd/Isa-Rd arm, whereas this number was 9, 15, and 38 patients in the VRd/Rd arm, respectively. MRD-neg, minimal residual disease–negative.

In these IMROZ analyses, we have evaluated MRD negativity in patients with VGPR or better, rather than only in patients with CR or better, because it provides more evidence on actual residual disease status in treated patients, in line with other reports.[12,23,25,26] Moreover, our findings along with other studies further support the Isa-VRd quadruplet regimen as standard of care for the frontline treatment of transplant-ineligible patients with NDMM.

All these findings on MRD dynamics in patients receiving Isa-VRd/Isa-Rd over 60 months suggest that assessing MRD negativity and MRD-negative CR at multiple time points supports treatment decisions, including selection, continuation, or discontinuation, in patients with NDMM. Understanding the potential benefits of prolonged treatment to manage disease control is of particular clinical relevance to MRD-positive patients; in this study, prolonged treatment with isatuximab-based quadruplet regimens resulted in some MRD-positive patients converting to MRD-negative status. This finding is supported by recent results from the IsKia study, in which after consolidation and before maintenance, light consolidation treatment was introduced. During light consolidation, a higher percentage of patients with MRD negativity at 10^{-5} became MRD negative at 10^{-6} with isatuximab-based regimens compared with control treatment.[27] Such an approach is supported by a phase 2 study demonstrating the feasibility of an NGS MRD-adapted strategy in NDMM. The strategy includes multiple MRD assessments at various steps after induction with an anti-CD38 antibody plus KRd, and selection of lenalidomide maintenance for MRD-negative patients and ASCT or KRd consolidation for MRD-positive patients.[28]

In conclusion, our landmark analyses on the extent of MRD negativity and MRD-negative CR benefit achieved at the end of the induction phase and during maintenance by patients treated with Isa-VRd/Isa-Rd compared with that by patients treated with VRd/Rd reinforce and extend the IMROZ primary study analyses.[10,11] Moreover, they further support the Isa-VRd quadruplet regimen as a standard of care for the frontline treatment of transplant-ineligible patients with NDMM.

Acknowledgments

The authors thank the participating patients and their caregivers, the study centers, and the investigators for their contributions to the study. The coordination of the development of this manuscript, facilitation of author discussion, and critical review were provided by Wendell Lamar Blackwell, Global Publications Lead of Scientific Communications at Sanofi. The authors also thank Umer Khan, Jothi Meenakshi, Jayesh Patil, and Sagar Sakat of Sanofi for programming/biostatistical support.

This study was funded by Sanofi. Medical writing support was provided by S. Mariani of Envision Pharma Group, funded by Sanofi.

Authorship

Contribution: R.Z.O., M.A.D., T.F., S.M., C.T., A.T.S., and P.M. contributed to the design of this study; and all authors contributed to the acquisition, analysis, and interpretation of the study data, critically reviewed the manuscript, and provided final approval for publication.

Conflict-of-interest disclosure: R.Z.O. reports consulting for AbbVie, Adaptive Biotechnologies, Biotheryx, Bristol Myers Squibb/Celgene, CARsgen, Cellcentric, GlaxoSmithKline, IASO Biotherapeutics, Karyopharm Therapeutics, Lytica Therapeutics, Meridian Therapeutics, Monte Rosa Therapeutics, Neoleukin Corporation, Oncopeptides, Pfizer, Regeneron, Sanofi, and Takeda; stocks in Asylia Therapeutics; and funding (to institution) from CARsgen, Exelixis, Heidelberg Pharma, and Sanofi. M.A.D. reports honoraria for participation in advisory boards and satellite symposia from Amgen, BeiGene, Bristol Myers Squibb, GlaxoSmithKline, Janssen, Menarini Stemline, Regeneron, Sanofi, and Takeda. T.I. reports grants or contracts from Alexion Pharmaceuticals, Bristol Myers Squibb, GlaxoSmithKline, Janssen, Pfizer, Prothena, Sanofi, and Takeda; and honoraria for lectures, presentations, or participation in speakers bureaus from Bristol Myers Squibb, Janssen, Ono, Sanofi, and Takeda. R.H. reports grants or contracts from Amgen, Bristol Myers Squibb, Celgene, Janssen, Novartis, and Takeda; consulting fees from AbbVie, Amgen, Bristol Myers Squibb, Celgene, Janssen, Novartis, PharmaMar, and Takeda; honoraria for lectures, presentations, and participation in speakers bureaus from Amgen, Bristol Myers Squibb, Celgene, Janssen, PharmaMar, and Takeda; support for attending meetings and/or travel from Amgen, Celgene, Janssen, and Takeda; and participation on a data safety monitoring board or advisory board for Amgen, Bristol Myers Squibb, GlaxoSmithKline, Janssen, Oncopeptides, Sanofi, and Takeda. I.S. reports consulting fees from Amgen, Bristol Myers Squibb, GlaxoSmithKline, Janssen-Cilag, Sanofi, and Takeda; honoraria for lectures, presentations, and participation in speakers bureaus from Amgen, Bristol Myers Squibb, Janssen-Cilag, Sanofi, and Takeda; support for attending meetings and/or travel from Janssen-Cilag, Sanofi, and Takeda; and participation on a data safety monitoring board or advisory board for Amgen, Janssen-Cilag, Sanofi, and Takeda. J.R.-J. reports speakers bureau participation for and honoraria from AstraZeneca, Celgene, Gilead, Janssen, Roche, Sanofi, and Takeda; and support for attending meetings and/or travel from Gilead, Roche, and Takeda. V.V. reports honoraria for lectures, presentations, and participation in speakers bureaus from AbbVie, Amgen, AstraZeneca, Bristol Myers Squibb, Janssen, MSD, Novartis, Pfizer, Roche, and Sanofi. B.B. reports honoraria from Amgen, GlaxoSmithKline, Janssen, Oncopeptides, Pfizer, and Takeda. T.J. reports research funding from Janssen and Sanofi; and honoraria from and consultancy/advisory roles with Bristol Myers Squibb, GlaxoSmithKline, Janssen, Pfizer, and Sanofi. H.G. reports grants or contracts from Amgen, Array BioPharma/Pfizer, Bristol Myers Squibb/Celgene, Chugai, Dietmar-Hopp-Foundation, Janssen, Johns Hopkins University, Mundipharma GmbH, and Sanofi; consulting fees from Adaptive Biotechnology, Amgen, Bristol Myers Squibb, Janssen, and Sanofi; honoraria from Amgen, Bristol Myers Squibb, Chugai, GlaxoSmithKline, Janssen, Novartis, Pfizer, and Sanofi; support for attending meetings and/or travel from Amgen, Bristol Myers Squibb, GlaxoSmithKline, Janssen, Novartis, Pfizer, and Sanofi; and research funding (to institution) from Bristol Myers Squibb/Celgene, GlycoMimetics Inc, GSK, Heidelberg Pharma, Hoffmann-La Roche, Incyte Corp, Janssen, Karyopharm, Millennium Pharmaceuticals Inc, Molecular Partners, MSD, MorphoSys AG, Novartis, Pfizer, Sanofi, and Takeda. T.M. reports research funding from Bristol Myers Squibb and Sanofi; and honoraria from GlaxoSmithKline, Pfizer, and Roche. M.M. reports a consultancy role with Bristol Myers Squibb, Janssen, Jazz, Pfizer, and Sanofi; honoraria from Adaptive, Amgen, Bristol Myers Squibb, GlaxoSmithKline, Janssen, Jazz, Novartis, Pfizer, Sanofi, Menarini Stemline, and Takeda; research funding from Jazz, Janssen, Pfizer, and Sanofi; and speakers bureau participation for Jazz, Janssen, Pfizer, and Sanofi. S.M., E.K., C.T., and A.T.S. are employed by Sanofi and may hold stock and/or stock options in the company. P.M. reports honoraria from and a consulting/advisory role with Amgen, Bristol Myers Squibb/Celgene, GlaxoSmithKline, Janssen, Pfizer, Sanofi, and Takeda. The remaining authors declare no competing financial interests.

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Footnotes

Submitted 4 June 2025; accepted 15 December 2025; prepublished online on *Blood* First Edition 27 February 2026. https://doi.org/10.1182/blood.2025030120.

Qualified researchers may request access to patient-level data and related study documents including the clinical study report, study protocol with any amendments, blank case report form, statistical analysis plan, and data set specifications. Patient-level data will be anonymized and study documents will be redacted to protect the privacy of the trial participants. Further details on Sanofi's data-sharing criteria, eligible studies, and process for requesting access can be found at https://www.vivli.org/.

The online version of this article contains a data supplement.

There is a *Blood* Commentary on this article in this issue.

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